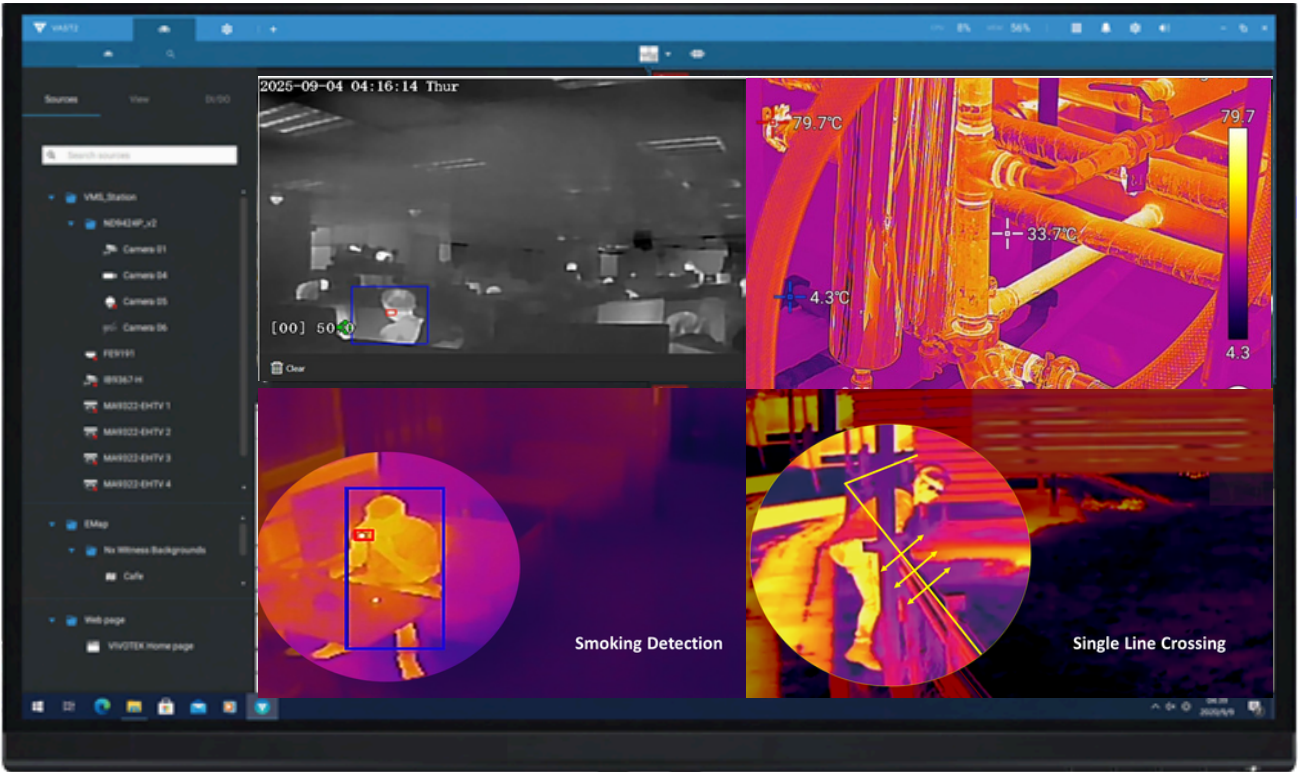


Thermal Bullet Cameras

Infrared Thermal Imaging



Temperature Measurement

Temperature Range	-20°C to 150°C (-4°F to 302°F)
Measurement Rule	Point, line, polygon (up to 20 rules in total)
Temperature Alarm	Threshold alarm, Temperature difference alarm, Temperature rise alarm, Section alarm
Accuracy	± 2°C / ±2%

AI Analytics

Compute Capability	SoC with built-in hardware deep learning accelerator (4T)
Application	Temperature measurement alarms (difference, threshold, section, rise), Early Fire Detection (Fire spot detection, Smoking detection) VCA (Intrusion, Single line crossing, Double line crossing, Loitering, Wrong way, Enter, Leave) Support VCA alarms triggering by person, car, person or car; Support temperature alarms triggering by ignoring person, car, person or car

Alarm and Event

Event Trigger	Motion detection, I/O alarm, Disk alarm, VCA alarm, Temperature alarm
Event Action	'Event notification via audio clip, digital output, email, FTP, SD card

V-series AI Camera

FD9189-V3 / FD9389-V3 / IB9389-V3 / FD9199 / FD9399 / IB9399



VIVOTEK's V-series AI cameras combine advanced intelligence with high-quality imaging, available in both 5MP and 8MP options. With AI-driven incident detection and real-time alerts, the cameras provide live-streaming capabilities to enhance deterrence. The user-friendly design reduces installation time and maximizes safety. This product series is the ideal solution for building a reliable and effective security system.

Features

● Hardware ● Software

5 / 8MP Resolution

AI Camera



40 Meters IR Illuminator



HDMI Output

Streams live camera footage

Simple Installation

Single (T10) Screw Mechanism and Interchangeable I/O Cable

TPM 2.0 Encryption Chip

Protects sensitive data and credentials

Trend-Micro™ IoT Security

Defend and Quarantine Vulnerability Protection



Vision Object Analytic

Identify people, vehicles and attributes



Smart VCA

Accurate detect incidents



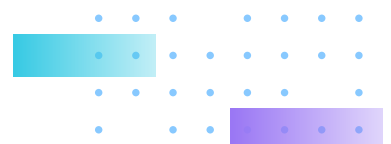
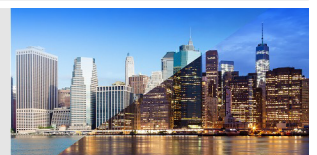
Deep Search

Available with supported platform



RealSight Engine

Clear Visibility of people in any lighting



Great Image Performance Powered by RealSight Engine

Clear visibility of people in any lighting

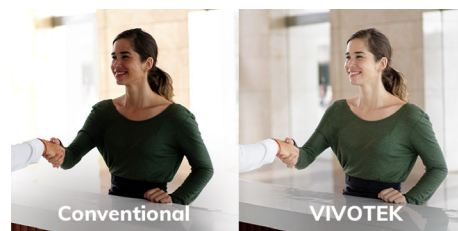
This feature ensures sharp, detailed capture of individuals, even in motion or challenging lighting conditions, without manual adjustments, the RealSight Engine automatically delivers clear images no matter day or night.



Clearly capture moving figures



See individuals without being affected by IR light reflections

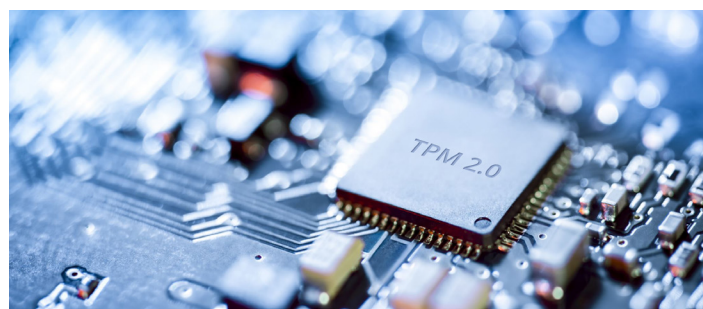
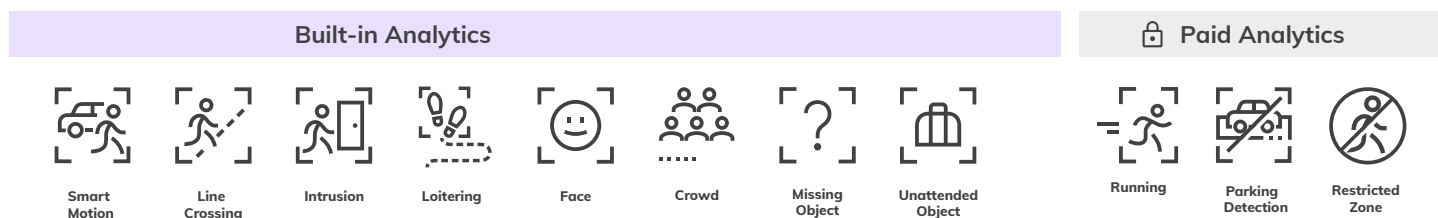


Clearly see portraits under backlight

Advanced Analytics

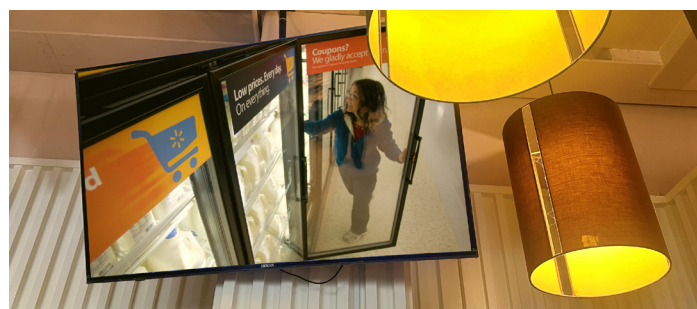
AI detection for accurate alarm

VIVOTEK Smart VCA offers precise object and attribute tracking, providing detailed video analytics for targeted monitoring and quick searches.



FIPS 140-2 Certified TPM 2.0 Chip

The TPM 2.0 chip provides hardware encryption to protect confidential and cryptographic data, prevent unauthorized access, and also ensures system integrity for long-term security.



HDMI output direct stream live footage

The camera is equipped with a built-in HDMI output port, allowing it to connect directly to the monitor via an HDMI cable. This setup eliminates the need for a video receiver to display surveillance footage, requires no additional wiring, and does not interfere with the existing decoration.

Build for Easy Installation

Simple installation

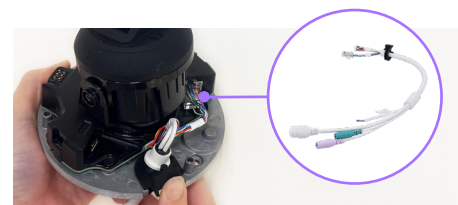
VIVOTEK's product design facilitates an easy and efficient installation process: only a single size of screwdriver is required to disassemble and install the camera device. Additionally, compatible accessories are provided to meet various installation requirements across different environments.



Effortless Cable Installation



Single (T10) Screw Mechanism



Interchangeable I/O Cable

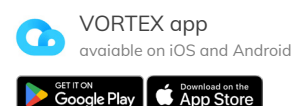
Complete Video Security with VIVOTEK Solution

Access and manage all devices and accounts through a single platform, providing accurate event monitoring and intuitive security system operation video investigation no matter where you are.



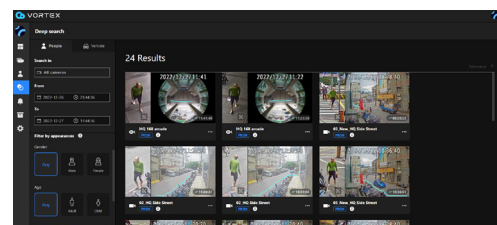
Instant notification to aware incident |

Integrated with VORTEX, key individuals can be promptly notified through mobile push on VORTEX app or email, enabling proactive deterrent measures and the prevention of violations.



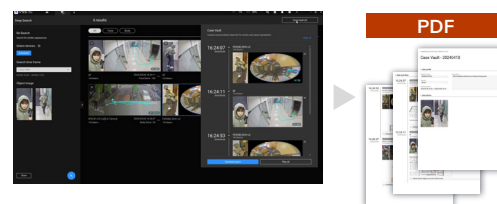
Best way to search for footage – Deep Search |

Effortlessly find evidence without the need to access numerous video footages by using Deep Search, which enables efficient searching for video clips based on various attributes. View snapshots of people and their paths, eliminating the need for video replays.

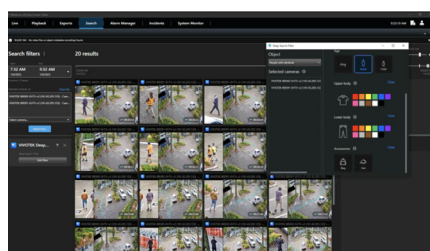


Generate case report and export video in organized |

Case Vault helps user to manage footage found through Deep Search. Organize clips and export detailed case reports straight from search results, making case management easier.

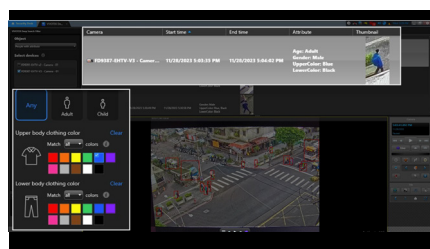


Unified experience on Milestone and Genetec



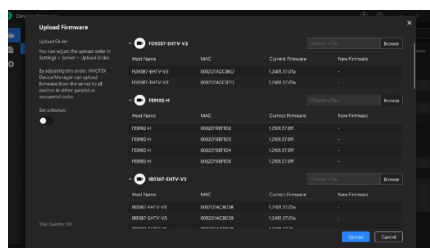
Milestone XProtect

Easily enhance your Milestone XProtect with VIVOTEK's AI cameras for instant search evidence by attributes.



Genetec Security Center

Optimize your Genetec Security Center with VIVOTEK's AI cameras for quick evidence searching through Deep Search and export search report.



VIVOTEK Device Manager – Batch configuration

Device Manager quickly configure your cameras and NVRs. With our batch configuration, IP allocation, HTTPS/IEEE certificates, and firmware updates have never been easier, ensuring the system is always up-to-date and streamlined.



Ordering Information

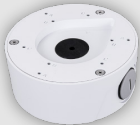
Models	FD9189-H-V3	FD9199-H	FD9389-EHV-V3	FD9389-EHTV-V3	FD9399-EHV	FD9399-EHTV	IB9389-EHV-V3	IB9389-EHTV-V3	IB9399-EHV	IB9399-EHTV
Form Factor	Indoor Fixed Dome		Outdoor Fixed Dome				Bullet			
Resolution	5MP	8MP	5MP	5MP	8MP	8MP	5MP	5MP	8MP	8MP
Lens	2.8mm	4.2mm	2.8mm	2.8-12mm	4.2mm	4.3-9.8mm	2.8mm	2.8-12mm	4.2mm	4.3-9.8mm
Aperture	F2.0	F1.6	F2.0	F1.6-3.3	F1.6	F1.6-2.6	F2.0	F1.6-3.3	F1.6	F1.6-2.6
IR illuminator	30m	30m	40m	40m	40m	40m	40m	40m	40m	40m
HDMI Output	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗
Certification	✗	✗	IP66, IK10, NEMA 4X	IP66, IK10, NEMA 4X	IP66, IK10, NEMA 4X	IP66, IK10, NEMA 4X	IP67, IK10	IP67, IK10	IP67, IK10	IP67, IK10

Accessories

Dome camera



AM-21A (V02)
L-shape Bracket



AM-71B (V01&V02)
Waterproof Conduit Box



AO-014
Audio & IO cable for indoor



AO-015
Audio & IO cable for outdoor

Bullet camera



AM-71C (V02)
Junction Box

Recommend Solution

Hybrid Cloud Solution

VORTEX Camera (Not within scope)

V-series AI Camera

V-series Embedded NVR

VORTEX BY VIVOTEK

On-Prem Solution

V-series AI Camera

V-series Recording Server

V-series Embedded NVR

VAST Security Station

Open Platform Solution

V-series AI Camera

milestone

Genetec

NxWitness

ONVIF® | GST



S Series

TB9333-E (3.5 mm)

Thermal Bullet Network Camera



Technical Specifications

System Information

CPU	Multimedia SoC (System-on-Chip)
Flash	16GB
RAM	1GB

Camera Specifications

Image Sensor	Dual sensor: Uncooled VOx Micro bolometer 1/2.7" Progressive CMOS
Max. Resolution	Thermal lens: Effective pixels: 256x192 Image scale up to: 704x576 Optical lens: 2880x1620
Lens Type	Fixed-focal (Thermal lens) Fixed-focal (optical lens)
Focal Length	Thermal lens: f = 3.5 mm Optical lens: f = 4 mm
Aperture	Thermal lens: F1.0 optical lens: F1.6
Iris Type	Fixed iris
Field of View	Thermal lens(3.5mm) 48°(Horizontal) 36° (Vertical) optical lens (4mm) 92° (Horizontal) 46°(Vertical)
Illuminators	Built-in IR Illuminator up to 35 meters; White light for strobe light warning
Pan/Tilt/Zoom Functionalities	ePTZ: 4x digital zoom

Storage	MicroSD/SDHC/SDXC card (256GB capacity supported)
Pixel Size	12 μ m
Spectral Range	8 ~ 14 μ m
Sensitivity	NETD < 50 mK

Video	
Compression	H.265, H.264, MJPEG
Maximum Frame Rate	Thermal: 25 fps @ 704x576 Optical: 25 fps @ 2880x1620
Maximum Streams	2 simultaneous streams each sensor
Main Stream	Thermal: 25fps@704x576 Optical: 25fps@2880x1620, 2560x1440, 2304x1296, 1920x1080, 1280x720
2nd Stream	Thermal: 25fps@704x576 , 352x240, 256x192 Optical: 25fps@704x576 , 640x480
Video Streaming	Adjustable resolution, CBR/VBR, Smart Encode
Image Settings	General settings video title and time stamp overlay, video orientation (flip, mirror, rotation) Thermal settings white balance, image adjustment (brightness, contrast, sharpness) Privacy mask Scheduled profile settings

Audio	
Audio Capability	Two-way audio
Compression	G.711, RAW_PCM
Interface	External microphone input, External line output, Built-in speaker

Network	
Users	Live viewing for up to 10 clients
Security	Access list, Password protection, user access log, user account management
Protocols	IPv4/IPv6, HTTP, RTSP/RTP/RTCP, TCP/UDP, DHCP, DNS, PPPOE, SMTP, HTTPS, 802.1x, QOS, NTP, TCP/IP, FTP, UPnP, DDNS, ICMP, IGMP
Interface	10 Base-T/100 Base-TX Ethernet (RJ-45) *It is highly recommended to use standard CAT5e & CAT6 cables which are compliant with the 3P/ETL standard.
ONVIF	Profile G, S, T supported, specification available at www.onvif.org

AI Analytics	
Compute Capability	SoC with built-in hardware deep learning accelerator (1T)

Application	Temperature measurement alarms (difference, threshold, section, rise), Early Fire Detection (Fire spot detection, Smoking detection, Smoke & flame detection) VCA (Intrusion, Single line crossing, Double line crossing, Loitering, Wrong way, Enter, Leave) Support VCA alarms triggering by person, car, person or car; Support temperature alarms triggering by ignoring person, car, person or car
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Solution Integration

Video Motion Detection	Smart motion detection
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Alarm and Event

Event Trigger	Motion detection, I/O alarm, Disk alarm, VCA alarm, Temperature alarm
Event Action	Event notification via audio clip, digital output, email, FTP, SD card; white light flashing and horn warning

General

Connectors	RJ-45 cable connector for 10/100 Mbps Network/PoE connection Audio input Audio output DC 12V power input AC 24V power input Digital input *2 Digital output *2 RS485
LED Indicator	System power and status indicator
Power Input	AC 24V, DC 12V, IEEE 802.3af PoE Class 3
Power Consumption	Max.12W
Dimensions	ø141.8 x 325 x 106.7mm
Weight	1960g
Certifications	EMC: CE (EN 55032, Class B), FCC (FCC Part 15 Subpart B), UKCA (BS EN 55032 Class B) Environment: IP66
Operating Temperature	Working Temperature: -40°C ~ 60°C (-40°F ~ 140°F)
Humidity	90%

System Requirements

Operating System	Microsoft Windows 10/8/7 Mac 10.12 (Chrome only)
Web Browser	Chrome 58.0 or above Internet Explorer 10/11
Other Players	VLC: 1.1.11 or above

Temperature Measurement

Temperature Range	-20°C to 150°C (-4°F to 302°F)
Measurement Rule	Point, line, polygon (up to 20 rules in total)
Temperature Alarm	Threshold alarm, Temperature difference alarm, Temperature rise alarm, Section alarm
Accuracy	± 2°C / ±2%

Included Accessories

Packing Contents	Quick installation guide, alignment sticker, screw pack, waterproof accessory bag, junction box, sunshield
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VCA Detection Range

VCA Detection Distance (Human: 1.8x0.5m)	30m (98.4ft)
VCA Detection Distance (Vehicle: 4x1.5m)	90m (295.3ft)

Temperature Measurement

Object: 0.2x0.2m	11.7m (38.4ft)
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Fire Detection

Object: 1x1m	208m (682ft)
Object: 0.2x0.2m	41.7m (136ft)
Smoking Detection	13m (42ft)

Johnson's Criteria (Thermal image)

Human 1.8x0.5m

Detection (1.5 pixels)	146m (479ft)
Recognition (6 pixels)	36m (118ft)
Identification (12 pixels)	18m (59ft)

Vehicle 4x1.5m

Detection (1.5 pixels)	447m (1466ft)
Recognition (6 pixels)	112m (367ft)
Identification (12 pixels)	56m (183ft)



S Series

TT9333-E (3.5 mm)

Thermal Turret Network Camera



Technical Specifications

System Information

CPU	Multimedia SoC (System-on-Chip)
Flash	16GB
RAM	1GB

Camera Specifications

Image Sensor	Dual sensor: Uncooled VOx Micro bolometer 1/2.7" Progressive CMOS
Max. Resolution	Thermal lens: Effective pixels: 256x192 Image scale up to: 704x576 Optical lens: 2560x1440
Lens Type	Fixed-focal (Thermal lens) Fixed-focal (optical lens)
Focal Length	Thermal lens: f = 3.5 mm optical lens: f = 4 mm
Aperture	Thermal lens: F1.0 optical lens: F1.6
Iris Type	Fixed iris
Field of View	Thermal lens(3.5mm) 48°(Horizontal) 36° (Vertical) optical lens (4mm) 92° (Horizontal) 46°(Vertical)
Illuminators	Built-in IR Illuminator up to 35 meters; White light for strobe light warning
Pan/Tilt/Zoom Functionalities	ePTZ: 4x digital zoom

Storage	MicroSD/SDHC/SDXC card (256GB capacity supported)
Pixel Size	12 μ m
Spectral Range	8 ~ 14 μ m
Sensitivity	NETD < 50 mK

Video	
Compression	H.265, H.264, MJPEG
Maximum Frame Rate	Thermal: 25 fps @ 704x576 Optical: 25 fps @ 2560x1440
Maximum Streams	2 simultaneous streams each sensor
Main Stream	Thermal: 25fps@704x576 Optical: 25fps@2560x1440, 2304x1296, 1920x1080, 1280x720
2nd Stream	Thermal: 25fps@704x576 , 352x240, 256x192 Optical: 25fps@704x576 , 640x480
Video Streaming	Adjustable resolution, CBR/VBR, Smart Encode
Image Settings	General settings video title and time stamp overlay, video orientation (flip, mirror, rotation) Thermal settings white balance, image adjustment (brightness, contrast, sharpness) Privacy mask Scheduled profile settings

Audio	
Audio Capability	Two-way audio
Compression	G.711, RAW_PCM
Interface	External microphone input External line output, Built-in microphone & speaker

Network	
Users	Live viewing for up to 10 clients
Security	Access list, Password protection, user access log, user account management
Protocols	IPv4/IPv6, HTTP, RTSP/RTP/RTCP, TCP/UDP, DHCP, DNS, PPPOE, SMTP, HTTPS, 802.1x, QOS, NTP, TCP/IP, FTP, UPnP, DDNS, ICMP, IGMP
Interface	10 Base-T/100 Base-TX Ethernet (RJ-45) *It is highly recommended to use standard CAT5e & CAT6 cables which are compliant with the 3P/ETL standard.
ONVIF	Profile G, S, T supported, specification available at www.onvif.org

AI Analytics	
Compute Capability	SoC with built-in hardware deep learning accelerator (1T)

Application	Temperature measurement alarms (difference, threshold, section, rise), Early Fire Detection (Fire spot detection, Smoking detection, Smoke & flame detection) VCA (Intrusion, Single line crossing, Double line crossing, Loitering, Wrong way, Enter, Leave) Support VCA alarms triggering by person, car, person or car; Support temperature alarms triggering by ignoring person, car, person or car
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Solution Integration	
Video Motion Detection	Smart motion detection

Alarm and Event	
Event Trigger	Motion detection, I/O alarm, Disk alarm, VCA alarm, Temperature alarm
Event Action	Event notification via audio clip, digital output, email, FTP, SD card; white light flashing and horn warning

General	
Connectors	RJ-45 cable connector for 10/100 Mbps Network/PoE connection Audio input Audio output DC 12V power input AC 24V power input Digital input *1 Digital output *1 RS485
LED Indicator	System power and status indicator
Power Input	AC 24V, DC 12V, IEEE 802.3af PoE Class 3
Power Consumption	Max.10W
Dimensions	φ140 × 117.8 mm
Weight	1040g
Certifications	EMC: CE (EN 55032, Class B), FCC (FCC Part 15 Subpart B), UKCA (BS EN 55032 Class B) Environment: IP66
Operating Temperature	Working Temperature: -40°C ~ 60°C (-40°F ~ 140°F)
Humidity	90%

System Requirements	
Operating System	Microsoft Windows 10/8/7 Mac 10.12 (Chrome only)
Web Browser	Chrome 58.0 or above Internet Explorer 10/11
Other Players	VLC: 1.1.11 or above

Temperature Measurement	
Temperature Range	-20°C to 150°C (-4°F to 302°F)
Measurement Rule	Point, line, polygon (up to 20 rules in total)
Temperature Alarm	Threshold alarm, Temperature difference alarm, Temperature rise alarm, Section alarm
Accuracy	± 2°C / ±2%

Included Accessories

Packing Contents	Quick installation guide, alignment sticker, screw pack, waterproof accessory bag
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VCA Detection Range

VCA Detection Distance (Human: 1.8x0.5m)	30m (98.4ft)
VCA Detection Distance (Vehicle: 4x1.5m)	90m (295.3ft)

Temperature Measurement

Object: 0.2x0.2m	11.7m (38.4ft)
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Fire Detection

Object: 1x1m	208m (682ft)
Object: 0.2x0.2m	41.7m (136ft)
Smoking Detection	13m (42ft)

Johnson's Criteria (Thermal image)

Human 1.8x0.5m

Detection (1.5 pixels)	146m (479ft)
Recognition (6 pixels)	36m (118ft)
Identification (12 pixels)	18m (59ft)

Vehicle 4x1.5m

Detection (1.5 pixels)	447m (1466ft)
Recognition (6 pixels)	112m (367ft)
Identification (12 pixels)	56m (183ft)



S Series

TB9332-E (9 mm)

Thermal Bullet Network Camera



Technical Specifications

System Information

CPU	Multimedia SoC (System-on-Chip)
Flash	16GB
RAM	1GB

Camera Specifications

Image Sensor	Uncooled VOx Micro bolometer
Max. Resolution	Effective pixels: 640x512 Image scale up to: 1280x1024
Lens Type	Fixed-focal (Thermal lens)
Focal Length	f = 9 mm
Aperture	F1.0
Iris Type	Fixed iris
Field of View	48° (Horizontal) 38° (Vertical)
Pan/Tilt/Zoom Functionalities	ePTZ: 4x digital zoom
Storage	MicroSD/SDHC/SDXC card (256GB capacity supported)
Pixel Size	12 μm
Spectral Range	8 ~ 14 μm
Sensitivity	NETD < 40 mK

Video

Compression	H.265, H.264
Maximum Frame Rate	25 fps @ 1280x1024

Maximum Streams	2 simultaneous streams
Main Stream	25fps @ 1280x1024, 1280x720, 640x512
2nd Stream	25fps @ 704x576 , 352x240,
Video Streaming	Adjustable resolution, CBR/VBR, Smart Encode
Image Settings	General settings video title and time stamp overlay, video orientation (flip, mirror, rotation)
	Thermal settings white balance, image adjustment (brightness, contrast, sharpness)
	Privacy mask Scheduled profile settings

Audio	
Audio Capability	Two-way audio
Compression	G.711, RAW_PCM
Interface	External microphone input, External line output

Network	
Users	Live viewing for up to 10 clients
Security	Access list, Password protection, user access log, user account management
Protocols	IPv4/IPv6, HTTP, RTSP/RTP/RTCP, TCP/UDP, DHCP, DNS, PPPOE, SMTP, HTTPS, 802.1x, QOS, NTP, TCP/IP, FTP, UPnP, DDNS, ICMP, IGMP
Interface	10 Base-T/100 Base-TX Ethernet (RJ-45) *It is highly recommended to use standard CAT5e & CAT6 cables which are compliant with the 3P/ETL standard.
ONVIF	Profile G, S, T supported, specification available at www.onvif.org

AI Analytics	
Compute Capability	SoC with built-in hardware deep learning accelerator (4T)
Application	Temperature measurement alarms (difference, threshold, section, rise), Early Fire Detection (Fire spot detection, Smoking detection)
	VCA (Intrusion, Single line crossing, Double line crossing, Loitering, Wrong way, Enter, Leave)
	Support VCA alarms triggering by person, car, person or car; Support temperature alarms triggering by ignoring person, car, person or car

Solution Integration	
Video Motion Detection	Smart motion detection

Alarm and Event	
Event Trigger	Motion detection, I/O alarm, Disk alarm, VCA alarm, Temperature alarm
Event Action	'Event notification via audio clip, digital output, email, FTP, SD card

General	
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Connectors	RJ-45 cable connector for 10/100 Mbps Network/PoE connection Audio input Audio output DC 12V power input AC 24V power input Digital input *2 Digital output *2 BNC RS485
LED Indicator	System power and status indicator
Power Input	AC 24V, DC 12V, IEEE 802.3af PoE Class 2
Power Consumption	Max. 5W
Dimensions	ø141.8 x 367.4 mm
Weight	2350g
Certifications	EMC: CE (EN 55032, Class B), FCC (FCC Part 15 Subpart B), UKCA (BS EN 55032 Class B) Environment: IP66
Operating Temperature	Working Temperature: -40°C ~ 60°C (-40°F ~ 140°F)
Humidity	90%

System Requirements	
Operating System	Microsoft Windows 10/8/7 Mac 10.12 (Chrome only)
Web Browser	Chrome 58.0 or above Internet Explorer 10/11
Other Players	VLC: 1.1.11 or above

Temperature Measurement	
Temperature Range	-20°C to 150°C (-4°F to 302°F)
Measurement Rule	Point, line, polygon (up to 20 rules in total)
Temperature Alarm	Threshold alarm, Temperature difference alarm, Temperature rise alarm, Section alarm
Accuracy	± 2°C / ±2%

Included Accessories	
Packing Contents	Quick installation guide, alignment sticker, screw pack, waterproof accessory bag, junction box, sunshield

VCA Detection Range	
VCA Detection Distance (Human: 1.8x0.5m)	47m (154.2ft)
VCA Detection Distance (Vehicle: 4x1.5m)	110m (360.9ft)

Temperature Measurement	
Object: 0.2x0.2m	30m (94.4 ft)

Fire Detection	
Object: 1x1m	375m (1230ft)

Object: 0.2x0.2m	75m (246ft)
Smoking Detection	32m (104ft)

Johnson's Criteria (Thermal image)	
Human 1.8x0.5m	
Detection (1.5 pixels)	375m (1230ft)
Recognition (6 pixels)	94m (308ft)
Identification (12 pixels)	47m (154ft)

Vehicle 4x1.5m	
Detection (1.5 pixels)	1150m (3772ft)
Recognition (6 pixels)	288m (944ft)
Identification (12 pixels)	144m (472ft)